# Experiment No. 5
# Name of experiment: Load Test of D.C Generator

1. ***Load Test of Separately Excited D.C Generator***
2. ***Load Test of self-excited D.C generator***

**1.1.Object:** To draw the external characteristics of a separately excited generator.

**1.2. Theory:** the external characteristics may be plotted for normal and for other value of excitation. Normal excitation is that value of the field current at which the machine gives rated voltage at its terminals when delivering rated current and running at rated speed.

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## 1.3. Procedure:

1. Connect the circuit as shown in figure 1.
2. Before starting, ensure that:
   **a.** there is no load on the generator ,i.e. switch is open.
   **b.** start the motor and adjust the speed at the rated value.
3. Increase the field current of the generator voltage till the induced voltage reaches the rated voltage.
4. Close the switch and increase the load current in different steps till it reaches rated current.
5. Record the value of all meters. At each step, keep the speed constant.

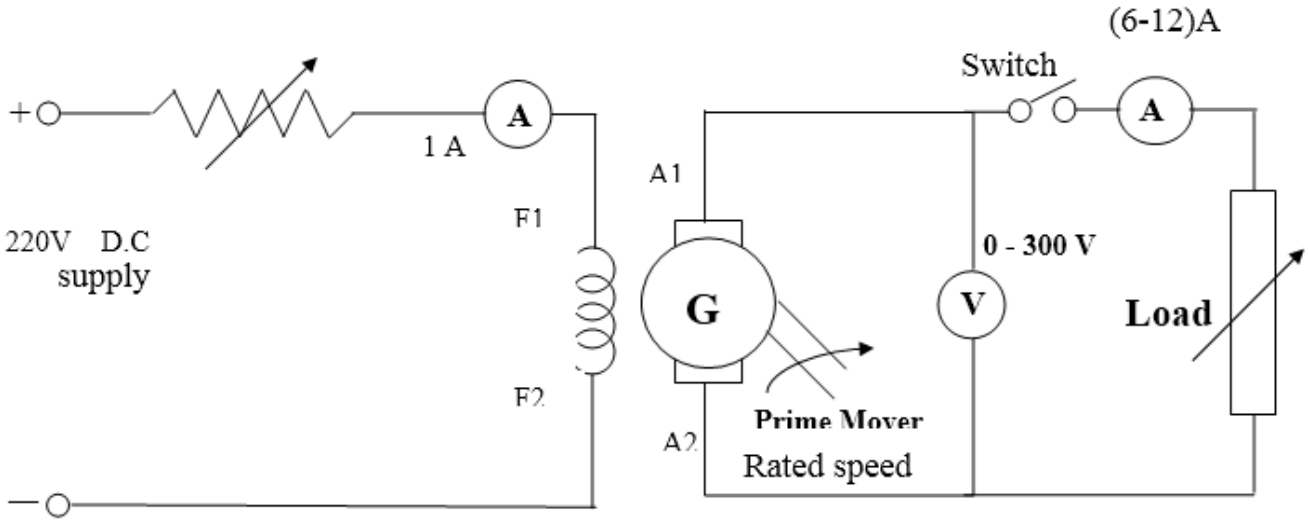



**Figure 1.**

| **IL**(load current) | **VL** (load voltage) | **Eg** (volt) | **If** | **Ia** |
|---|---|---|---|---|
| 0 | | | | |
| 1 | | | | |
| 1.5 | | | | |
| 2 | | | | |
| 2.5 | | | | |
| 3 | | | | |

**2.1 Object:** To draw the external characteristics of a self-excited generator.

**2.2. Theory:** the external characteristics may be plotted for normal and for other value of excitation. Normal excitation is that value of the field current at which the machine gives rated voltage at its terminals when delivering rated current and running at rated speed.

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## 2.3. Procedure:

1. Connect the circuit as shown in figure 2.
2. Before starting, ensure that:
   **a.** there is no load on the generator ,i.e. switch is open.
   **b.** start the motor and adjust the speed at the rated value.
3. Increase the field current of the generator voltage till the induced voltage reaches the rated voltage.
4. Close the switch and increase the load current in different steps till it reaches rated current.
5. Record the value of all meters. At each step, keep the speed constant.

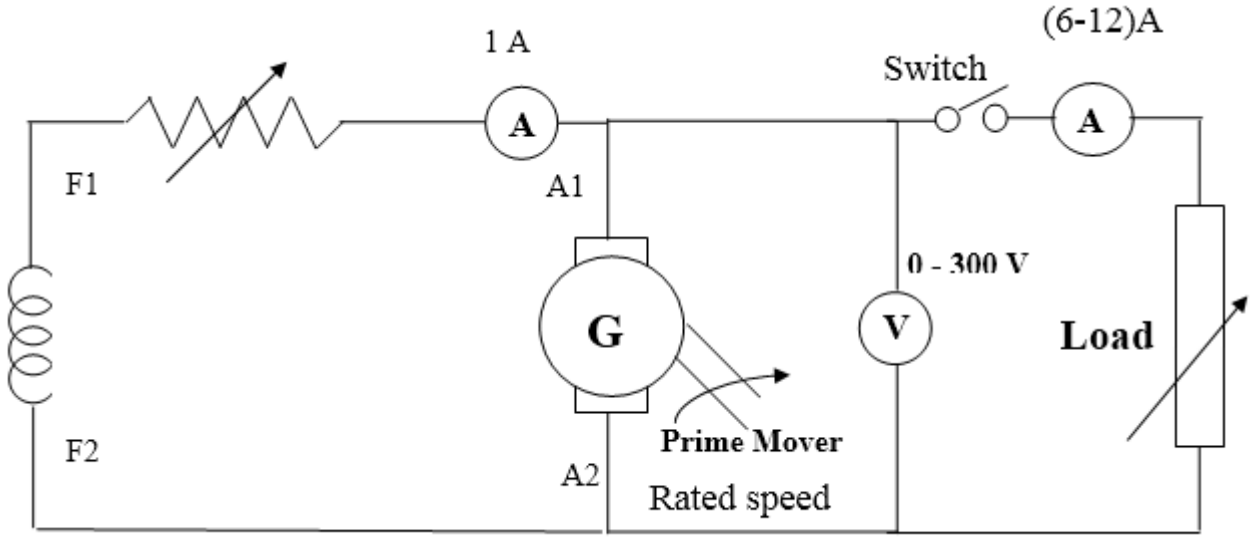



**Figure 2.**

| **IL**(load current) | **VL** (load voltage) | **Eg** (volt) | **If** | **Ia** |
|---|---|---|---|---|
| 0 | | | | |
| 1 | | | | |
| 1.5 | | | | |
| 2 | | | | |
| 2.5 | | | | |
| 3 | | | | |

## 1.4. Report (separately exited D.C generator):

1. Plot the external characteristics for self-excited generator.
2. If the resistance of armature is (4Ω) calculate the voltage drop at each steep.
3. Draw the internal characteristics of self-excited D.C generator.
4. Discuss your result briefly.
5. What are the reasons of fall of terminal voltage of a D.C shunt generator?
6. What happens if load resistance is decreased? Discuss briefly.

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## 2.4. Report (self-exited D.C generator):

1- Plot the external characteristics for self-excited generator.
2- If the resistance of armature is (4Ω) calculate the voltage drop at each steep.
3- Draw the internal characteristics of self-excited D.C generator.
4- Discuss your result briefly.
5- What are the reasons of fall of terminal voltage of a D.C shunt generator?
6- What happens if load resistance is decreased? Discuss briefly.